Amir Mirzai Golpayegani

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Education

M.Sc. in Computer Science Sep 2022 – Feb 2026
University of Calgary, Calgary, AB, Canada
Relevant Courses: Computer Animation (A+), Modelling for Computer Graphics (A)
Thesis: Research in indexing schemes and continuous data representation in graphics and geospatial contexts
GPA: 3.940/4.0

B.Sc. in Computer Engineering Sep 2016 – Feb 2021
Sharif University of Technology, Tehran, Iran
GPA: 3.69/4.0 (17.37/20)

Publications

1. **Mirzai Golpayegani, Amir**, M. Hasan, and F. F. Samavati, "Real-time multiresolution management of spatiotemporal earth observation data using dggs," *Remote Sensing*, vol. 17, no. 4, p. 570, 2025 doi:10.3390/rs17040570

Technical Skills

Programming Languages: Python, C++
Scientific Computing / Image Processing: NumPy, SciPy, Pandas, GDAL, SNAP, OpenGL, GLSL, data visualization
Software Engineering: Git, Docker, REST APIs, PostgreSQL, Linux, debugging, testing
Machine Learning: PyTorch, TensorFlow, Scikit-learn
Cloud / Platforms: AWS, GitHub Actions

Work Experience

Graphics Research Assistant Sep 2022 – Feb 2026
University of Calgary / BigGeo, Calgary, Canada

- Designed and implemented high-performance software components in C++ for large-scale image processing and spatiotemporal data analysis.
- Built automated frameworks and workflows for ingesting, encoding, compressing, and retrieving multiresolution scientific image data.
- Extended an existing DGGS/GDAL-based codebase with emphasis on correctness, maintainability, performance, and reusable software architecture.
- Developed GPU-accelerated visualization and rendering components using OpenGL/GLSL to support analysis and inspection workflows.
- Collaborated closely with researchers to translate algorithmic ideas into reliable, reproducible software tools.

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz, Tehran, Iran

- Developed production backend services and REST APIs in Python using Django.
- Implemented scalable backend components supporting NLP and speech-related applications.
- Diagnosed and resolved system issues using logging, profiling, and unit testing to improve reliability and maintainability.

Selected Projects

DEM Super-Resolution Framework May 2024 – Feb 2026
Python, PyTorch, CNNs, Scientific Imaging, Data Processing, OpenGL

- Designed and trained a PyTorch-based residual framework for image super-resolution on large-scale elevation data, with emphasis on reconstructing fine structural detail.
- Built automated data ingestion and preprocessing pipelines for global-scale training data generation and quality control.

- Developed 3D visualization and evaluation tools to inspect model outputs and support iterative model improvement.

Full-Stack Health and Wellness Tracker Jul 2025 – Present
Node.js, Express.js, PostgreSQL, JWT, Docker, GitHub Actions

- Building a secure backend system with authentication, activity logging, and AI-assisted recommendation features.
- Using PostgreSQL for persistent storage, JWT/bcrypt for authentication, and Docker for reproducible deployment.
- Automating build and deployment workflows through GitHub Actions.

Research Experience

Spatiotemporal Data Encoding and Storage Using Discrete Global Grids Sep 2022 – Feb 2026
University of Calgary *Supervisor: Dr. Faramarz Samavati*

- Researched efficient storage and retrieval methods for large time-varying imaging and Earth observation datasets.
- Developed multiresolution spatial representations for compression and scalable access.
- Investigated continuous temporal representations for time-series approximation and analysis.

Bachelor's Thesis: Persian Text Normalizer for NLP Oct 2016 – Feb 2021
Sharif University of Technology *Supervisor: Dr. Hossein Sameti*

- Developed a Python toolkit to standardize Persian text for downstream NLP pipelines.

Teaching Experience

Teaching Assistant, Developing Big Data Applications Dec 2024 – Apr 2025
University of Calgary *Instructor: Dr. Tyler Bonnell*
Mentored students in designing and deploying cloud-based solutions using AWS for large-scale data processing and storage, including hands-on guidance in debugging, optimizing workflows, and applying best practices for reliability and scalability.

Teaching Assistant, Working with Data at Scale Dec 2023 – Apr 2024
University of Calgary *Instructor: Dr. Katie Ovens*
Supported students in large-scale data processing, data pipeline design, and practical techniques for handling and analyzing high-volume datasets.

Teaching Assistant, Database Management Systems Dec 2022 – Apr 2023
University of Calgary *Instructor: Dr. Reda Alhaj*
Assisted students with SQL, database design, and course projects, and graded assignments and exams with attention to correctness and software quality.

Languages

English: Full Professional Proficiency (TOEFL 115)
Persian: Native Proficiency

Certificates

Supervised Machine Learning: Regression and Classification [certificate] Oct 2024
DeepLearning.AI and Stanford University, Coursera

Applied Machine Learning in Python [certificate] Jul 2020
University of Michigan, Coursera

Achievements

Ranked in the Top 0.001% of Konkour Participants May 2016
17th Nationwide, University Entrance Exam, Iran